AGENTIC AI BUILDING BLOCKS
AWS BEDROCK · SA DISCOVERY

You don't have a model problem. You have a discovery problem.

Anthropic, Amazon, Meta, Mistral, Cohere — the model landscape is overwhelming.
The answer isn't picking the "best" model. It's asking the right questions first.

Choosing the Right Bedrock Model: An SA-Led Discovery

PART I · THE PROBLEM

The Manual Claims Triage Nobody Wants to Do

Insurance claims handlers repeat the same cognitive workflow hundreds of times a day. Every step is manual. Every decision carries risk.

Receive	Read	Check	Lookup	Decide	Document
Claim arrives	Docs & photos	Policy lookup	Coverage details	Approve/Deny	Audit reasoning

⚠ False Approvals

Paying fraudulent claims — direct financial loss

⚠ False Rejections

Angry customers, potential lawsuits, brand damage

DISCOVERY · REQUIREMENTS

Six Requirements That Define the Solution

Before touching a single model, the SA-led discovery surfaces what actually matters. These aren't features — they're constraints that eliminate 80% of wrong choices.

Long Document Processing

Text PDFs + damage photos — multimodal input

Multi-System Integration

Policy DB, S3, fraud API, workflow system

High Accuracy Decisions

Wrong decisions = fraud payouts or lawsuits

Explainable Reasoning

Audit trail for regulatory compliance

Vision Capabilities

Assess property damage severity from photos

Sub-5s Latency

Handlers can't wait — speed is non-negotiable

"Start with your problem, not the models."

The Bedrock Model Landscape

Multi-provider access — each family brings distinct strengths

A Anthropic Claude

Opus 4.6

Most intelligent — enterprise agents

PREMIUM

Sonnet 4.6

95-99% of Opus at 5× lower cost

RECOMMENDED

Haiku 4.5

Hybrid reasoning, near-instant

FAST

N Amazon Nova

Nova 2 Pro

Beats Sonnet 4.5 on 10/16 benchmarks

MID-TIER

Nova 2 Lite

Beats Haiku 4.5 on 13/15, 1M ctx

BUDGET

Sonic / Omni

Voice & text+image generation

SPECIALTY

M Meta Llama

Scout 17B

Multimodal, 10M token context

MID-TIER

Maverick 17B

128 experts MoE, beats GPT-4o

MID-TIER

Mi Mistral

Large 3

675B params, 256k ctx, agentic

MID-TIER

Ministral 3B

Edge-optimized with vision

EDGE

The 4-Dimension Evaluation Framework

🧠

01

Capability

Can it handle your task complexity?

Complex policy reasoning with ambiguous language. Multi-clause interpretation. Nuanced judgment.

→ Sonnet 4.6, Opus 4.6, Nova 2 Pro

⏱️

02

Latency

Is it fast enough?

Under 5 seconds for standard claims. Sonnet 4.6 and Nova 2 Pro deliver within budget.

→ Sonnet 4.6, Nova 2 Pro (medium speed)

💰

03

Cost

Does it fit your budget?

Sonnet 4.6 delivers 95-99% of Opus at 5× lower cost. Multi-model cascade for savings.

→ Multi-model cascade: 60-70% savings

🔧

04

Features

Does it support what you need?

Vision for damage photos. Tool use for APIs. Structured output. Sonnet 4.6 excels at all three.

→ Sonnet 4.6 (best all-round)

AI AGENT ARCHITECTURE

Key Implementation Patterns

Four progressive patterns for building intelligent AI agents — from basic invocation to fully autonomous multi-model pipelines

1 2 3 4

PATTERN 1

Basic Model Invocation

- Connect to Bedrock Runtime
- Specify which model to use
- Send your prompt
- Receive the response
- Each model has slightly different request formats

Foundation

PATTERN 2

Multi-Model Testing

- Test same prompts across different models
- Measure accuracy, latency, cost
- Make data-driven decisions
- Documentation shows how to handle different model formats

Evaluation

PATTERN 3

Tool Use (Converse API)

- Define your tools as JSON schemas
- Model decides when to call tools
- You execute the actual API calls
- Return results to model
- Model synthesizes final answer
- Fully autonomous orchestration

Agentic

PATTERN 4

Multi-Model Pipeline

- Different models for different stages
- Route requests based on complexity
- Optimize cost and performance
- Can reduce costs significantly

Advanced

★ HERO ARCHITECTURE

The Multi-Model Claims Pipeline

Use the right model for each stage.

STAGE 1

Document Extraction

Nova 2 Lite

Fast & economical parsing

→

STAGE 2

Image Analysis

Sonnet 4.6

Analyze damage photos.
Excellent vision - only when images present

→

STAGE 3

Policy Reasoning

Sonnet 4.6

Complex coverage analysis.
Multi-step reasoning.
97-99% of Opus capability. High accuracy.

→

STAGE 4

Decision Generation

Sonnet 4.6

Approve / Deny / Escalate.
Structured output.
Audit Trail.

💡 **Right model per task = optimized cost + maximum accuracy**

Cost reduction vs. single premium model: **60–70%**

Right model for each task · Sonnet 4.6 handles 80-90% of use cases · Reserve Opus for edge cases only

PART III · PRODUCTION

From POC to Production-Ready

Resilience

- Cross-region inference profiles — automatic failover
- Retry with exponential backoff
- Fallback to faster models on timeout
- Escalate to human on system errors

Inference Profiles

- Cross-Region: AWS-managed, auto-route
- Application: Tag by customer — granular cost
- Cost Explorer integration with budgets
- Multi-tenant SaaS billing solved

Compliance

- Data processed in-region, never leaves
- Geography-scoped profiles (US/EU)
- VPC endpoints for private connectivity
- CloudTrail audit logging

Model Cascading

Light models for simple tasks, powerful models only where needed

Intelligent Routing

Bedrock auto-routes to most cost-effective model maintaining quality

Provisioned Throughput

Commit to capacity for high-volume — lower per-token cost at scale

CALL TO ACTION

Start Simple. Test Thoroughly. Then Optimize.

1

Prepare

FOUNDATION

✓ Setup Bedrock Permissions · ✓ Prepare representative claims dataset with edge cases

2

Compare

VALIDATION

✓ Test across models · ✓ Compare accuracy, latency, cost · ✓ Build POC · ✓ Pilot customer testing

3

Ship

SCALE

✓ Production deployment · ✓ Set up inference profiles · ✓ Implement monitoring · ✓ Scale to additional customers